



Science and
Technology
Facilities Council

63mm Tile Testing

Ben Thompson

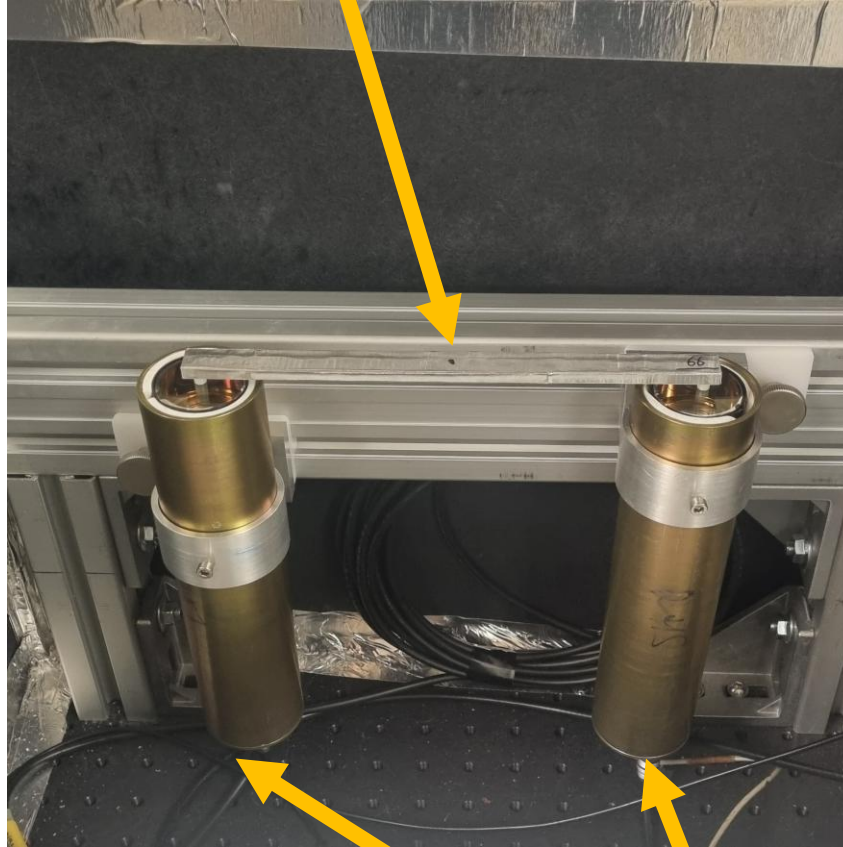


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Tile Testing Setup:

- Place scintillating tile onto two PMTs and Couple with optical Grease.
- Align the source over the top of the tile.
- When radiation hits the tiles, it produces light.
- Measure light produced by the tile on the PMTs
- This gives us a metric of how bright the tiles are.

Scintillating tile here



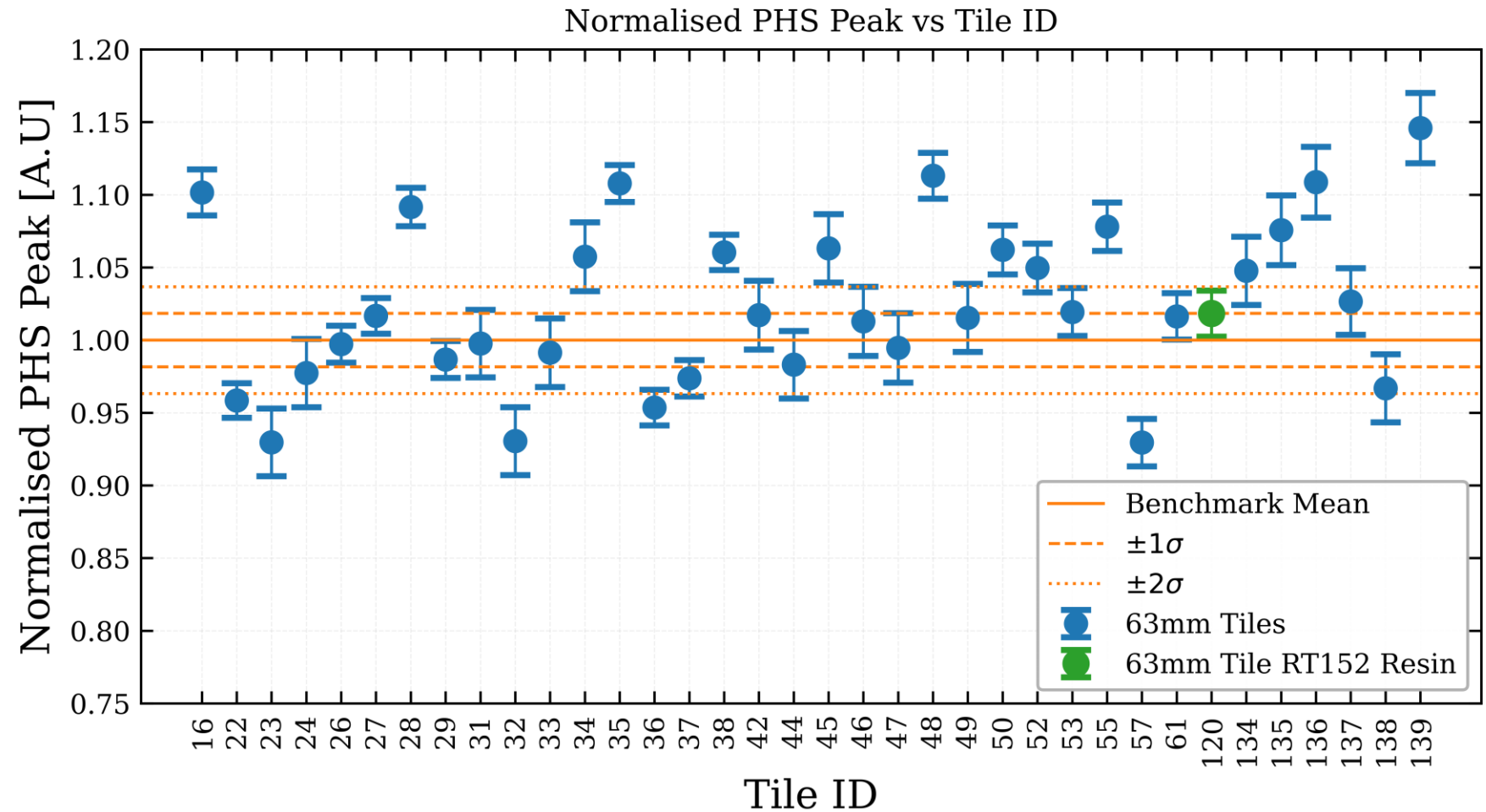
Radioactive source Here



PMTs are here

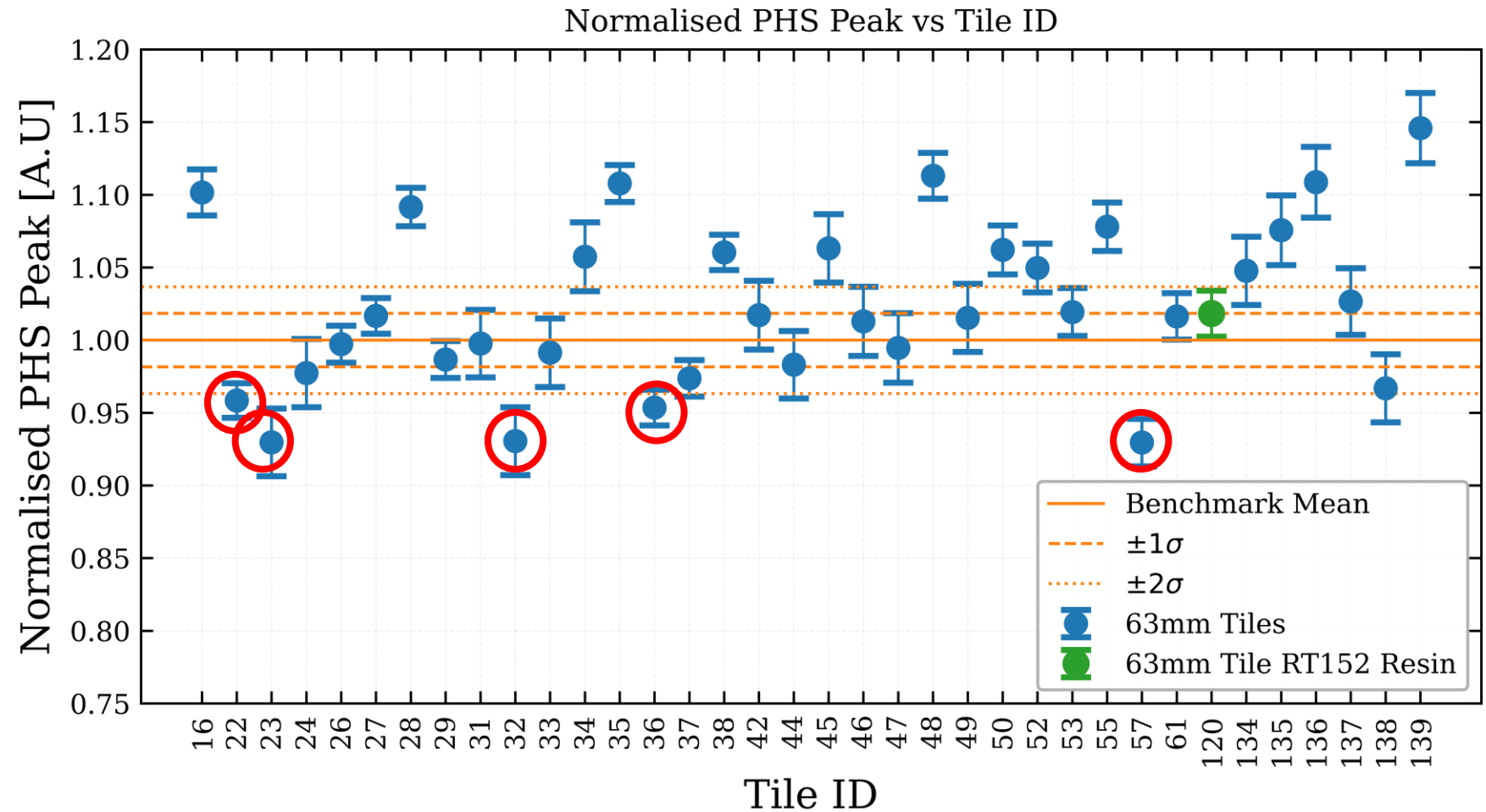
63mm Tile Testing

- 36 63mm Tiles were tested
- Of the 36 tiles, 5 were found to be below the Acceptance Threshold
- These tiles were investigated and compared to the production teams notes
- The majority have issues with the studs, so a t-test was done to see if there was a statistical difference between tiles with and without stud faults noted in production notes
- p-value of 0.697, no statistical difference @ $p \leq 0.05$



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ID	Resin	Production Notes
22	RAL71	Resin topped up, tiny bubble in each stud, one low divot
23	RAL71	Resin topped up, bubble in each stud
32	RAL71	Large bubbles at bottom of each leg
36	RAL71	Tiny nick on tile
57	RAL71	1 loose stud
134*	RAL71	Semi chilled resin, scuff on top, tiny bubbles in 1 stud, hairs present



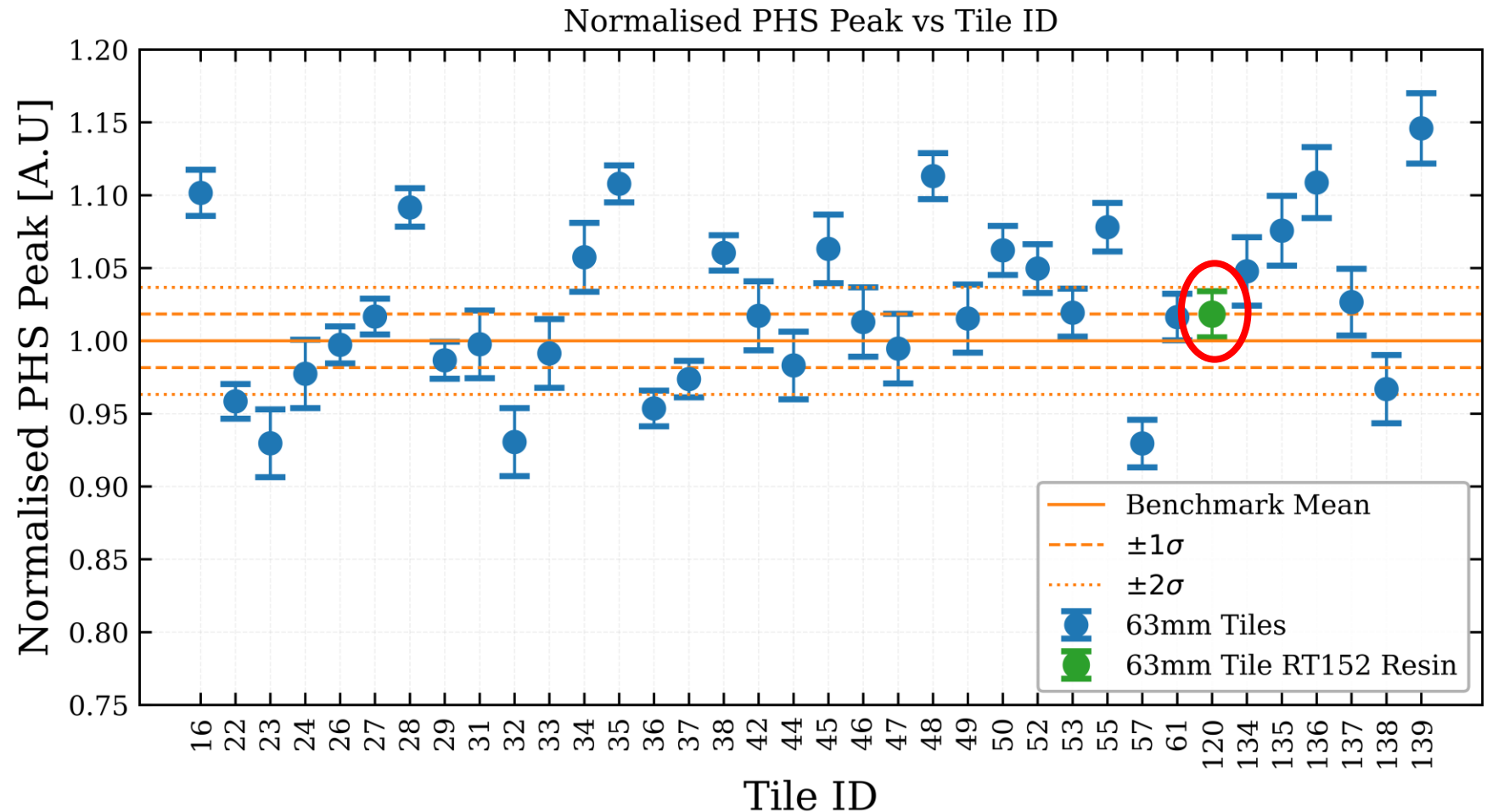
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RT152

- RA171a Resin was deforming when curing at high temperatures
- Wanted to investigate if RT152 Resin can perform within the acceptance threshold as it cures at 23°C and so doesn't encounter the deformation issues
- Tiles ID 120, 119 and 118 are made using this resin
- 118 and 119 were tested on the beamline and confirmed to be of suitable light output!
- This means that we can use this resin for longer length tiles.



Announcement :My Talk will be rescheduled

- Due to unforeseen circumstances involving the availability of my academic supervisor, the meeting will have to be rescheduled
- Currently reorganising, will update as soon as possible.
- Plan is to keep the time and location the same, but the potential dates are:
 - 6th August
 - 13th August
 - 14 August
- Sorry for the inconvenience!





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Thanks



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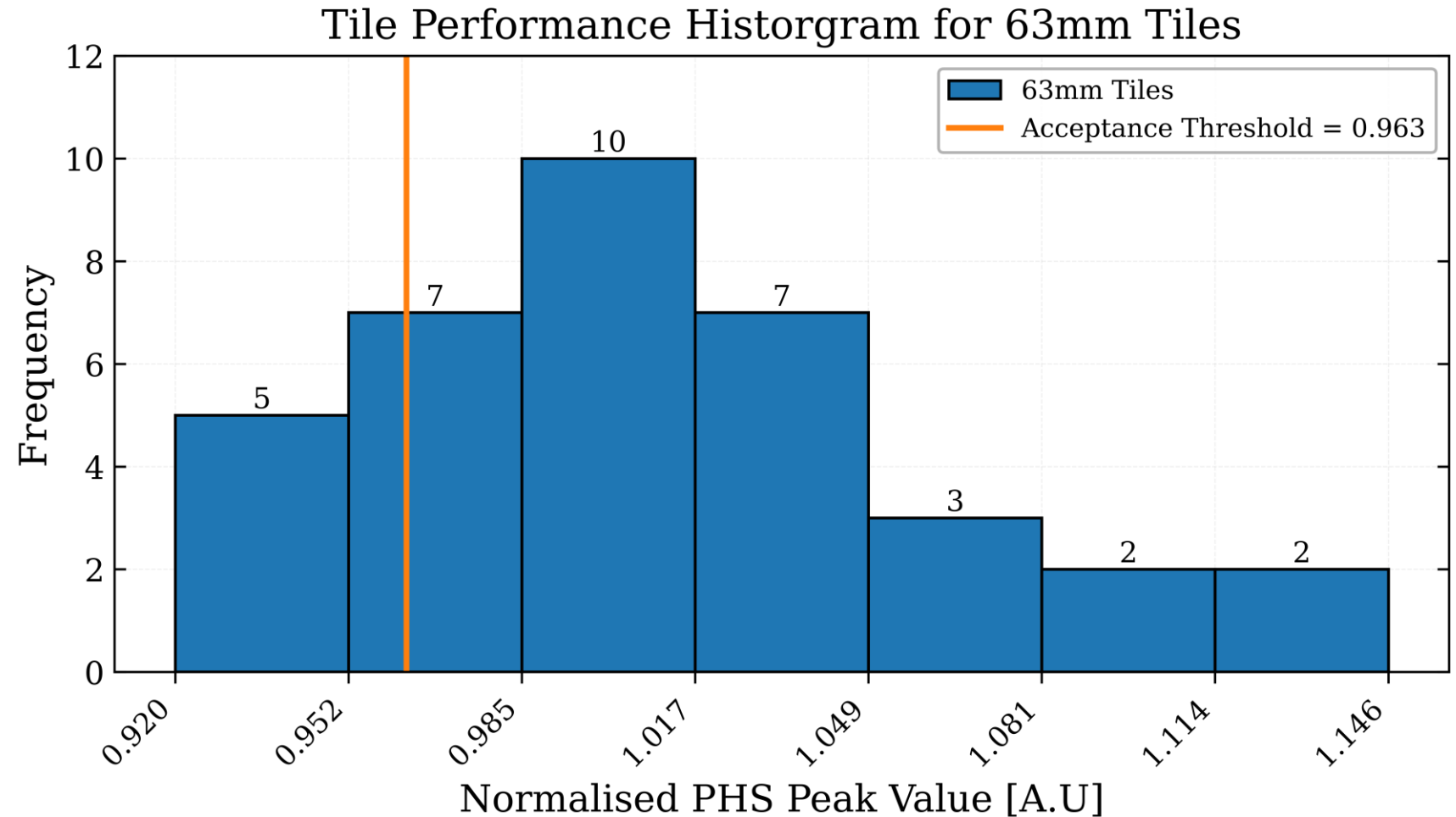
Bonus Slides



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63mm Tile Testing

- Histogram shows a positive skew gaussian distribution
- Suggests we have some randomness in the tile production process
- Cause likely human error as they are all hand made



Why the T-Test

